REAL-TIME ULTRASOUND PROBE MOTION TRACKING SYSTEM

End Term Report

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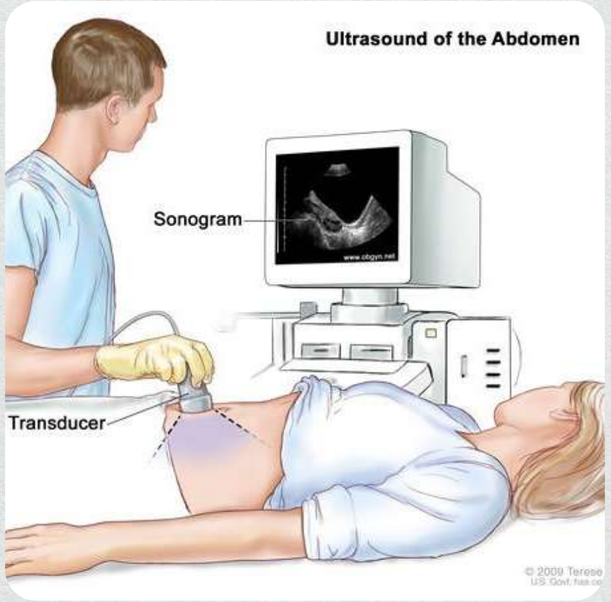
WHY DO WE NEED AN ULTRASOUND PROBE TRACKING SYSTEM?

1. Ultrasound is one of the **safest imaging modalities**, offering high **portability,** cost-effectiveness and **real-time** imaging capability

2. Ultrasound imaging is highly **operator-dependent** and lacks standardized acquisition protocols, as **image quality depends on probe movement**, positioning, and selection of the scanning region.

3. This variability **limits** the broader adoption and expansion of ultrasound as a reliable imaging modality.



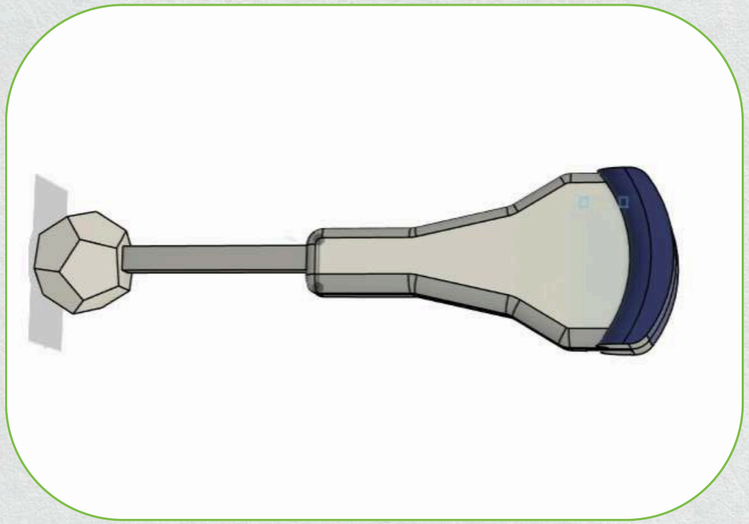
AIM:

- Our project focuses on building a **6 DOF ultrasound probe tracking system** that can provide information of the position, orientation and trajectory of the ultrasound probe in real-time.

- The approach is to **mount light-weight sensor modules** like IMU and optical markers and ESP32 inside the probe and a camera outside at a distance. The data feeds for the camera and IMU can be fused to give accurate pose and location of the probe.

METHODOLOGY

- **DESIGN:**
  - Designed and 3d printed a **hollow probe** with the design features of an actual ultrasound probe. Printed it in parts and connected them together via a connector.
  - A **dodecahedron** model is also printed and attached to the probe body which houses the aruco markers (one on each face of the polyhedron)



METHODOLOGY

- HARDWARE
  - **BNO085 IMU:** The 9DoF IMU with inbuilt sensor fusion is mounted inside the probe body.
  - **ESP32:** The ESP serially sends the orientation and position data from the IMU to edge processor via a USB cable. Wired mode used to reduce latency.
  - **Camera:** A USB camera provides secondary probe pose and position estimation using ArUco markers.

Composite Sensor	Maximum Data rates (Hz)
Gyro rotation Vector	1000
Rotation Vector	400
Gaming Rotation Vector	400
Geomagnetic Rotation Vector	90
Gravity	400
Linear Acceleration	400
Accelerometer	500
Gyroscope	400
Magnetometer	100

Figure 6-16: Maximum sensor rates

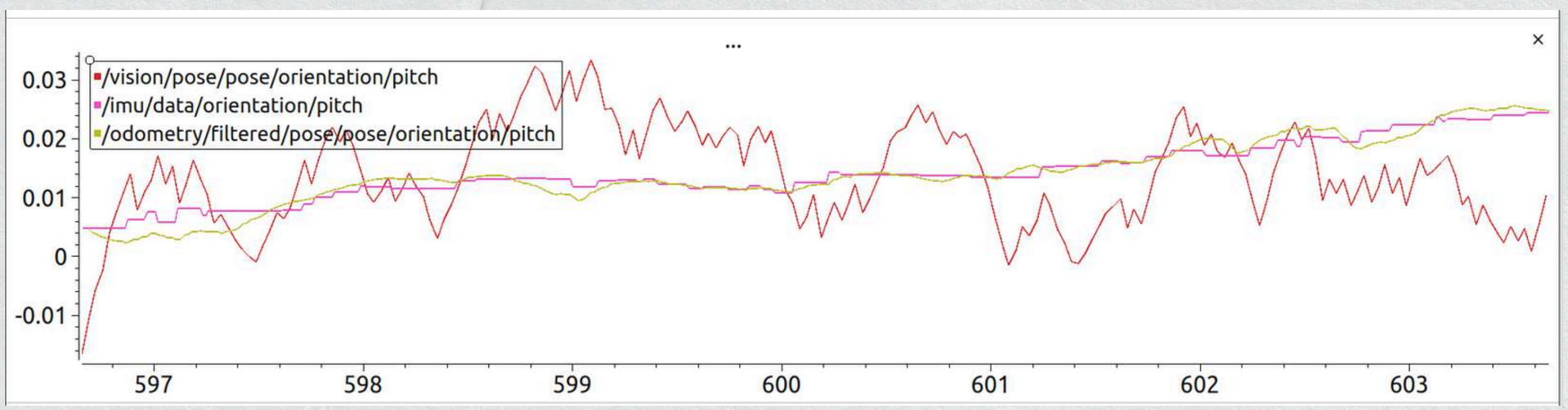
METHODOLOGY

- **CAD Design:**

- **Final Hardware prototype:**
  Probe with mounted hardware

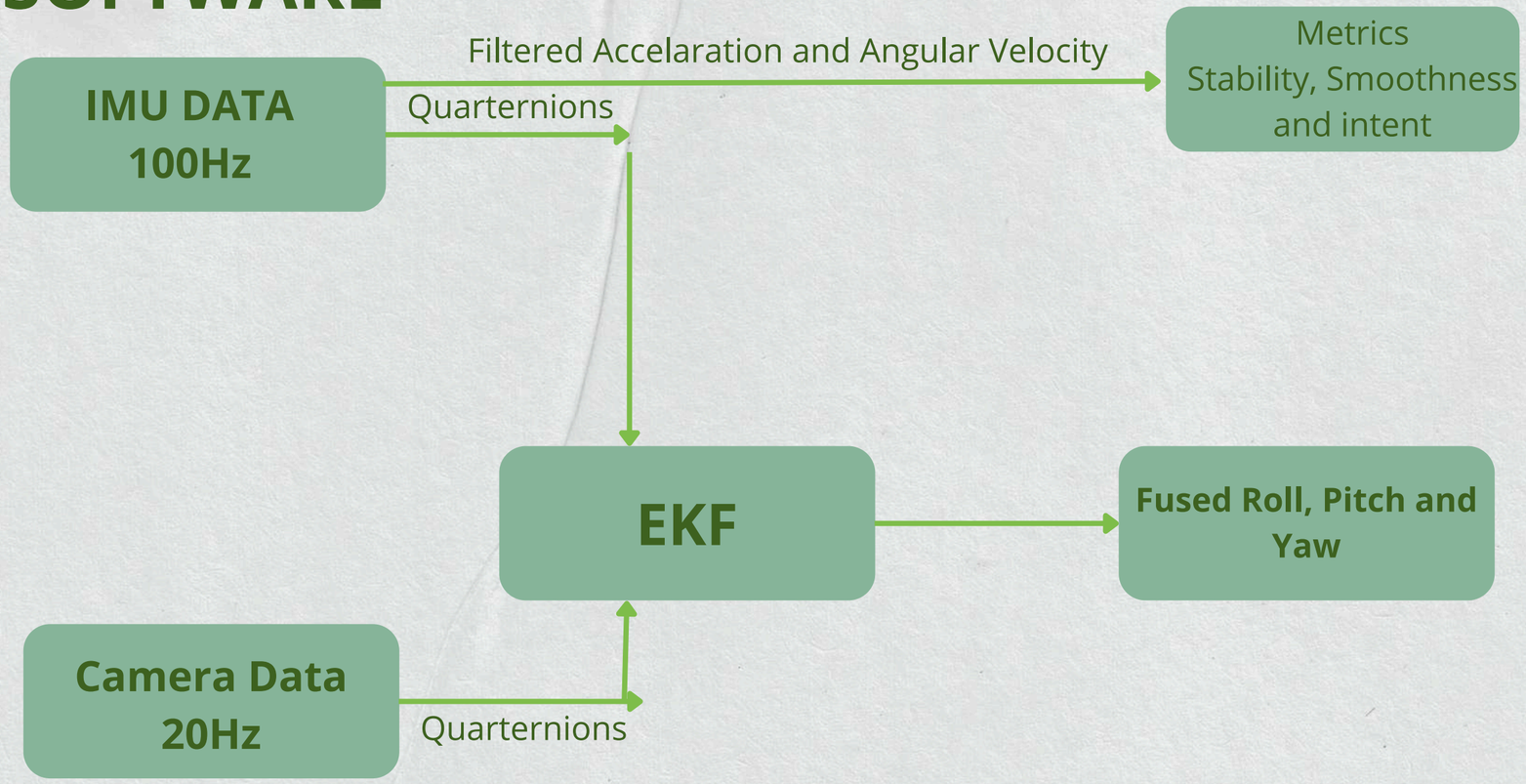
METHODOLOGY

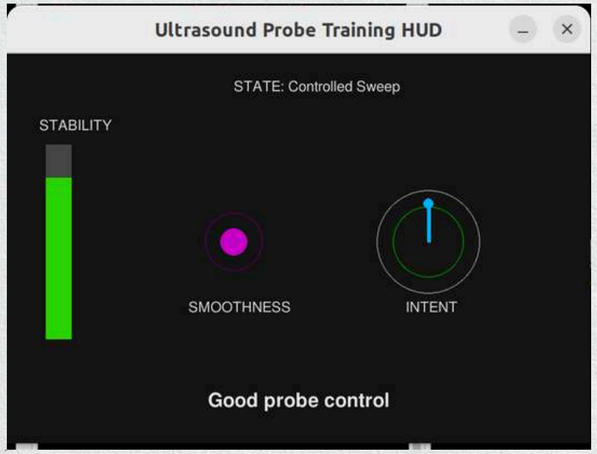
- **SOFTWARE:**
  - We are running Extended Kalman Filter (EKF) sensor fusion on the IMU and camera data streams in real-time.
  - The probe trajectory along with its pose as the output of the EKF are displayed using a python library Matplotlib.



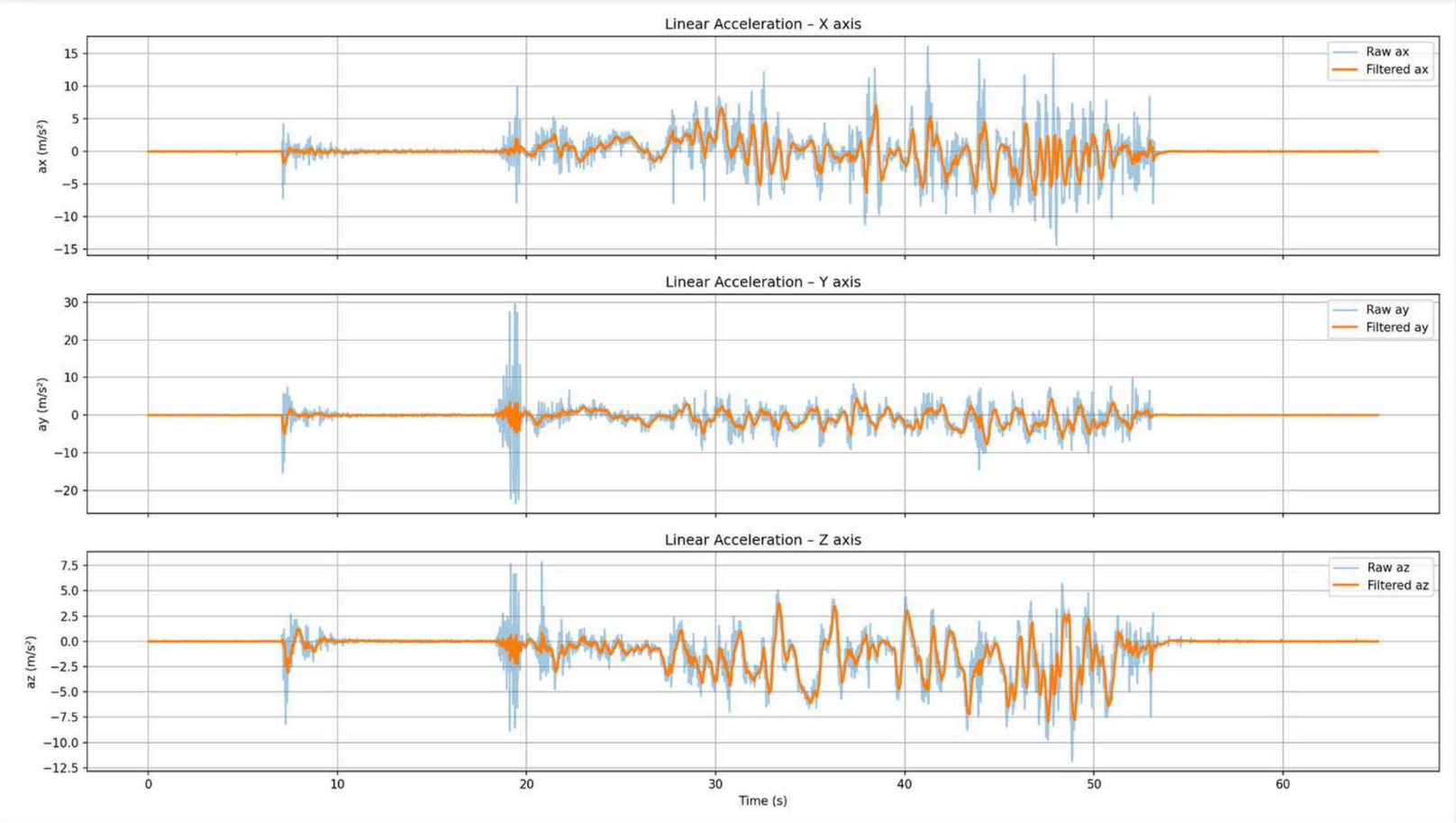
METHODOLOGY

SOFTWARE



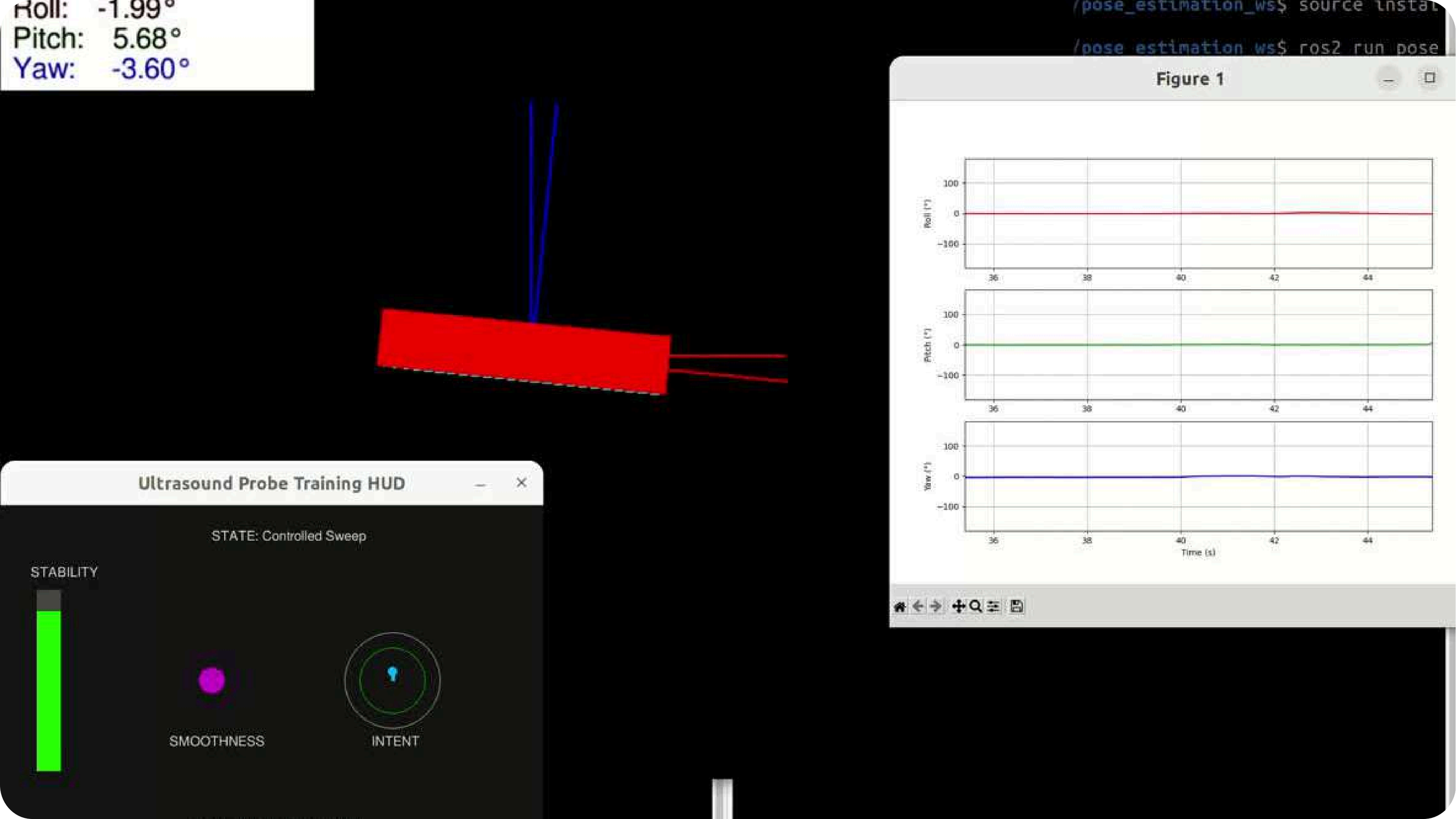


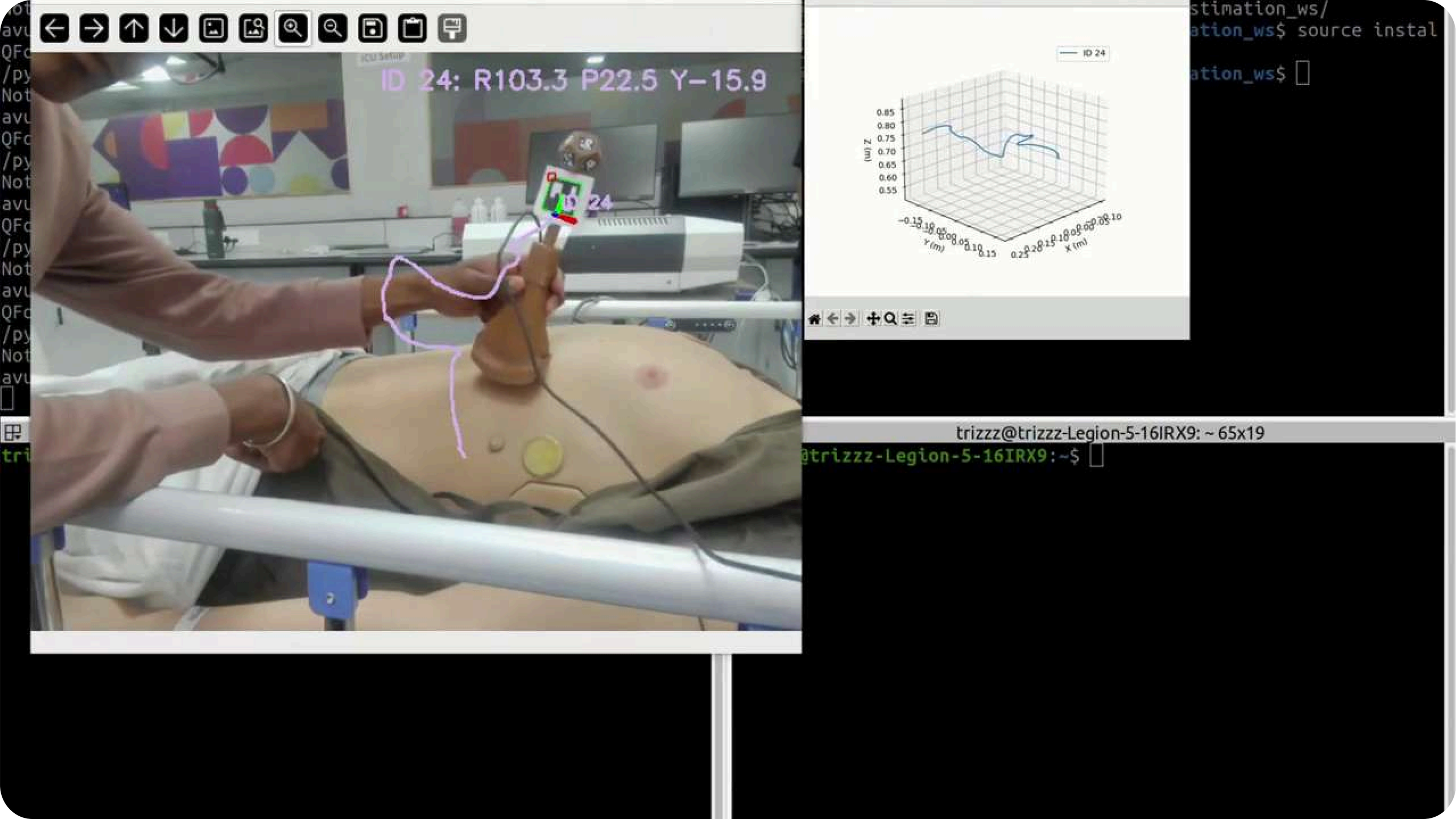
Data fed to metrics calculator



Cleaned IMU Accelaration Data using Kalman Filter to remove Noise and Bias

DEMO VIDEOS


Roll: -1.99°
Pitch: 5.68°
Yaw: -3.60°
/pose_estimation_ws$ source install
/pose_estimation_ws$ ros2 run pose
Figure 1
Roll (°)
Pitch (°)
Yaw (°)
100
0
−100
36
38
40
42
44
Time (s)
Ultrasound Probe Training HUD
STATE: Controlled Sweep
STABILITY
SMOOTHNESS
INTENT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APPLICATIONS

Robotic Ultrasound

- Closed-Loop: Feeds pose data to robotic arms for autonomous scans.
- Remote Sync: Replicates a doctor's hand motion via tele-operation.

Image-Guided Procedures

- Virtual Trajectory: Overlays needle paths for "pre-entry" planning.
- 3D Targeting: Pins precise coordinates for biopsy guidance.

Fusion with CT/MRI

- Live Slicing: Extracts CT/MRI slices that match probe pose.
- Dual Overlay: Merges real-time US with high-res anatomical data.

Ultrasound Training

- Path Scoring: Compares student motion to expert benchmarks.
- Steadiness Metrics: Quantifies hand stability and efficiency.

FUTURE SCOPE

- Developing a visualization pipeline to render the patient model in 3D, with the probe model moving according to the EKF-estimated pose.
- Integrating **touch sensors** into the probe to detect probe–surface contact and prevent **false detections.**
- Incorporating a **haptic feedback mechanism** to simulate realistic tactile sensations during ultrasound scanning training.
- GPU Implementation for faster processing and real-time performance.
- Validating the complete end-to-end pipeline

THANK YOU